TabuLLM: AI Benchmark Report

1. Executive Summary: Consolidated Audit Results

This report consolidates the results of the TabuLLM data augmentation pipeline. The objective is to audit the fidelity, diversity, and privacy of the synthetic data generated via LLM inference. Below are the averaged performance indicators across all processed datasets in this session.

Audit Indicator	Global Session Mean (± Std)
Synthetic Fidelity (JSD)	85.7 ± 2.28
Correlation Stability	89.5 ± 1.02
Distribution Consistency	87.3 ± 1.85
Entropy Preservation	88.3 ± 1.10
Mean Z Score Drift	0.1 ± 0.02
Categorical Fidelity	0.0 ± 0.00
Privacy Risk	100.0 ± 0.00
Data Density	100.0 ± 0.00
Coverage Score	0.0 ± 0.00
Feature Accuracy	89.5 ± 3.20
Novelty Score	0.0 ± 0.00

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Dataset Audit: heart_failure_clinical_records.csv

1. Dataset Portrait

Attribute	Value
Rows	299
Columns	13
Numeric features	13
Categorical features	0
Missing cells (%)	0.0%

2. Generation Strategy & LLM Config [1]

Parameter	Configuration
Algorithm (*12)	gpt-4o-mini
Folds (*13)	10
Context % (*14)	20%
Generation % (*15)	100%
Join context (*16)	True

3. Statistical Significance & Data Drift

4. Quality & Fidelity Metrics (Mean ± Std)

Metric Category	Aggregated Score
Synthetic Fidelity (JSD) (*1)	85.7 ± 2.28
Correlation Stability (*2)	89.5 ± 1.02
Distribution Consistency (*3)	87.3 ± 1.85
Entropy Preservation (*4)	88.3 ± 1.10
Mean Z Score Drift (*5)	0.1 ± 0.02
Categorical Fidelity (*6)	0.0 ± 0.00
Privacy Risk (*7)	100.0 ± 0.00
Data Density (*8)	100.0 ± 0.00
Coverage Score (*9)	0.0 ± 0.00
Feature Accuracy (*10)	89.5 ± 3.20
Novelty Score (*11)	0.0 ± 0.00

*[1] (*12) Algorithm: Contextual constraint or model parameter.*

*[2] (*13) Folds: Contextual constraint or model parameter.*

*[3] (*14) Context %: Contextual constraint or model parameter.*

*[4] (*15) Generation %: Contextual constraint or model parameter.*

*[5] (*16) Join context: Contextual constraint or model parameter.*

[6] Synthetic Fidelity (JSD): Overall distribution similarity.
[7] Correlation Stability: Multivariate relationships preserved.
[8] Distribution Consistency: Median feature similarity.
[9] Entropy Preservation: Information diversity maintained.
[10] Mean Z Score Drift: Standardized distribution shift.
[11] Categorical Fidelity: Category distribution similarity.
[12] Privacy Risk: Synthetic rows too close to originals.
[13] Data Density: Non-missing cells in augmented data.
[14] Coverage Score: Synthetic coverage of original space.
[15] Feature Accuracy: Mean feature similarity.
[16] Novelty Score: Distance ratio to nearest neighbors.

6. Downstream ML Utility: Comparative Analysis

Model: XGB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.835	N/A	0.835	0	
F1	0.7547418227770806	N/A	0.7547418227770806	0	

Model: GB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.8300000000000001	N/A	0.8300000000000001	0	
F1	0.7541632435801531	N/A	0.7532401666570763	-0.00092307692307692 64	

Model: DT

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7816666666666666	N/A	0.78	-0.00166666666666663	
F1	0.6698202237325381	N/A	0.6691945550978049	-0.00062566863473331 18	

Model: SVM

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6333333333333334	N/A	0.6333333333333334	0	
F1	N/A	N/A	N/A	0	

Model: LR

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.805	N/A	0.805	0	
F1	0.6894302795561282	N/A	0.6894302795561282	0	

Model: AB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.825	N/A	0.825	0	
F1	0.741740477144242				

7

N/A	0.741740477144242

7

0	

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Model: NB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7499999999999999	N/A	0.7499999999999999	0	
F1	0.5667171896628354	N/A	0.5667171896628354	0	

Model: KNN

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6133333333333335	N/A	0.6133333333333335	0	
F1	0.007142857142857143	N/A	0.007142857142857143	0	

Model: MLP

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.5	N/A	0.5383333333333333	0.03833333333333334	
F1	0.2403307790715618	N/A	0.31994842767591763	0.07961764860435581	

Model: RF

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.825	N/A	0.8400000000000001	0.01499999999999999	
F1	0.7305531467275165	N/A	0.7583047118634634	0.02775156513594699	

Appendix: Benchmark Data Trace

Legend: [Blue] = Real Context | [Green] = Generated Synthetic | [Yellow] = Real Training (Remaining) | [Grey] = Real Test Baseline

A. Baseline: Original Dataset (Full Record Sample)

#	age	anaemia	creatinine_phosphoki	diabetes	ejection_fractio	high_blood_pressu	platelets	serum_creatinine	serum_sodium	sex
1	75.0	0.0	582.0	0.0	20.0	1.0	265000.0	1.9	130.0	1.0
2	55.0	0.0	7861.0	0.0	38.0	0.0	263358.03	1.1	136.0	1.0
3	65.0	0.0	146.0	0.0	20.0	0.0	162000.0	1.3	129.0	1.0
4	50.0	1.0	111.0	0.0	20.0	0.0	210000.0	1.9	137.0	1.0
5	65.0	1.0	160.0	1.0	20.0	0.0	327000.0	2.7	116.0	0.0
6	90.0	1.0	47.0	0.0	40.0	1.0	204000.0	2.1	132.0	1.0
7	75.0	1.0	246.0	0.0	15.0	0.0	127000.0	1.2	137.0	1.0
8	60.0	1.0	315.0	1.0	60.0	0.0	454000.0	1.1	131.0	1.0
9	65.0	0.0	157.0	0.0	65.0	0.0	263358.03	1.5	138.0	0.0
10	80.0	1.0	123.0	0.0	35.0	1.0	388000.0	9.4	133.0	1.0
11	75.0	1.0	81.0	0.0	38.0	1.0	368000.0	4.0	131.0	1.0
12	62.0	0.0	231.0	0.0	25.0	1.0	253000.0	0.9	140.0	1.0
13	45.0	1.0	981.0	0.0	30.0	0.0	136000.0	1.1	137.0	1.0
14	50.0	1.0	168.0	0.0	38.0	1.0	276000.0	1.1	137.0	1.0
15	49.0	1.0	80.0	0.0	30.0	1.0	427000.0	1.0	138.0	0.0
16	82.0	1.0	379.0	0.0	50.0	0.0	47000.0	1.3	136.0	1.0
17	87.0	1.0	149.0	0.0	38.0	0.0	262000.0	0.9	140.0	1.0
18	45.0	0.0	582.0	0.0	14.0	0.0	166000.0	0.8	127.0	1.0
19	70.0	1.0	125.0	0.0	25.0	1.0	237000.0	1.0	140.0	0.0
20	48.0	1.0	582.0	1.0	55.0	0.0	87000.0	1.9	121.0	0.0

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B.1 Audit Trace: Fold 1 | Ctx: 20% | Gen: 100%

#	age	anaemia	creatinine_phosphoki	diabetes	ejection_fractio	high_blood_pressu	platelets	serum_creatinine	serum_sodium	sex
1	60.0	0.0	582.0	0.0	40.0	0.0	217000.0	3.7	134.0	1.0
2	70.0	0.0	571.0	1.0	45.0	1.0	185000.0	1.2	139.0	1.0
3	60.0	0.0	235.0	1.0	38.0	0.0	329000.0	3.0	142.0	0.0
4	50.0	1.0	298.0	0.0	35.0	0.0	362000.0	0.9	140.0	1.0
5	65.0	1.0	335.0	0.0	35.0	1.0	235000.0	0.8	136.0	0.0
6	70.0	1.0	171.0	0.0	60.0	1.0	176000.0	1.1	145.0	1.0
7	45.0	1.0	66.0	1.0	25.0	0.0	233000.0	0.8	135.0	1.0
8	70.0	0.0	161.0	0.0	25.0	0.0	244000.0	1.2	142.0	0.0
9	58.0	1.0	400.0	0.0	40.0	0.0	164000.0	1.0	139.0	0.0
10	55.0	0.0	572.0	1.0	35.0	0.0	231000.0	0.8	143.0	0.0

B.2 Audit Trace: Fold 2 | Ctx: 20% | Gen: 100%

#	age	anaemia	creatinine_phosphoki	diabetes	ejection_fractio	high_blood_pressu	platelets	serum_creatinine	serum_sodium	sex
1	50.0	1.0	1051.0	1.0	30.0	0.0	232000.0	0.7	136.0	0.0
2	50.0	0.0	582.0	0.0	50.0	0.0	153000.0	0.6	134.0	0.0
3	45.0	0.0	582.0	0.0	35.0	0.0	385000.0	1.0	145.0	1.0
4	70.0	0.0	582.0	1.0	38.0	0.0	25100.0	1.1	140.0	1.0
5	80.0	0.0	805.0	0.0	38.0	0.0	263358.03	1.1	134.0	1.0
6	62.0	0.0	231.0	0.0	25.0	1.0	253000.0	0.9	140.0	1.0
7	60.0	0.0	2656.0	1.0	30.0	0.0	305000.0	2.3	137.0	1.0
8	45.0	1.0	130.0	0.0	35.0	0.0	174000.0	0.8	139.0	1.0
9	45.0	0.0	582.0	1.0	55.0	0.0	543000.0	1.0	132.0	0.0
10	65.0	1.0	59.0	1.0	60.0	0.0	172000.0	0.9	137.0	0.0

B.3 Audit Trace: Fold 3 | Ctx: 20% | Gen: 100%

#	age	anaemia	creatinine_phosphoki	diabetes	ejection_fractio	high_blood_pressu	platelets	serum_creatinine	serum_sodium	sex
1	55.0	0.0	748.0	0.0	45.0	0.0	263000.0	1.3	137.0	1.0
2	80.0	0.0	776.0	1.0	38.0	1.0	192000.0	1.3	135.0	0.0
3	49.0	0.0	789.0	0.0	20.0	1.0	319000.0	1.1	136.0	1.0
4	75.0	1.0	246.0	0.0	15.0	0.0	127000.0	1.2	137.0	1.0
5	73.0	1.0	1185.0	0.0	40.0	1.0	220000.0	0.9	141.0	0.0
6	46.0	0.0	168.0	1.0	17.0	1.0	271000.0	2.1	124.0	0.0
7	50.0	0.0	582.0	1.0	38.0	0.0	310000.0	1.9	135.0	1.0
8	79.0	1.0	55.0	0.0	50.0	1.0	172000.0	1.8	133.0	1.0
9	65.0	1.0	160.0	1.0	20.0	0.0	327000.0	2.7	116.0	0.0
10	70.0	0.0	92.0	0.0	60.0	1.0	317000.0	0.8	140.0	0.0

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B.4 Audit Trace: Fold 4 | Ctx: 20% | Gen: 100%

#	age	anaemia	creatinine_phosphoki	diabetes	ejection_fractio	high_blood_pressu	platelets	serum_creatinine	serum_sodium	sex
1	45.0	0.0	2442.0	1.0	30.0	0.0	334000.0	1.1	139.0	1.0
2	75.0	0.0	99.0	0.0	38.0	1.0	224000.0	2.5	134.0	1.0
3	60.0	1.0	2281.0	1.0	40.0	0.0	283000.0	1.0	141.0	0.0
4	65.0	1.0	128.0	1.0	30.0	1.0	297000.0	1.6	136.0	0.0
5	46.0	1.0	291.0	0.0	35.0	0.0	348000.0	0.9	140.0	0.0
6	50.0	0.0	124.0	1.0	30.0	1.0	153000.0	1.2	136.0	0.0
7	48.0	1.0	131.0	1.0	30.0	1.0	244000.0	1.6	130.0	0.0
8	95.0	1.0	371.0	0.0	30.0	0.0	461000.0	2.0	132.0	1.0
9	45.0	1.0	981.0	0.0	30.0	0.0	136000.0	1.1	137.0	1.0
10	50.0	0.0	582.0	1.0	38.0	0.0	310000.0	1.9	135.0	1.0

B.5 Audit Trace: Fold 5 | Ctx: 20% | Gen: 100%

#	age	anaemia	creatinine_phosphoki	diabetes	ejection_fractio	high_blood_pressu	platelets	serum_creatinine	serum_sodium	sex
1	73.0	1.0	1185.0	0.0	40.0	1.0	220000.0	0.9	141.0	0.0
2	50.0	0.0	115.0	0.0	45.0	1.0	184000.0	0.9	134.0	1.0
3	55.0	0.0	1820.0	0.0	38.0	0.0	270000.0	1.2	139.0	0.0
4	60.0	0.0	3964.0	1.0	62.0	0.0	263358.03	6.8	146.0	0.0
5	72.0	0.0	364.0	1.0	20.0	1.0	254000.0	1.3	136.0	1.0
6	60.0	1.0	754.0	1.0	40.0	1.0	328000.0	1.2	126.0	1.0
7	78.0	0.0	224.0	0.0	50.0	0.0	481000.0	1.4	138.0	1.0
8	70.0	0.0	582.0	0.0	20.0	1.0	263358.03	1.83	134.0	1.0
9	61.0	0.0	248.0	0.0	30.0	1.0	267000.0	0.7	136.0	1.0
10	40.0	0.0	478.0	1.0	30.0	0.0	303000.0	0.9	136.0	1.0

B.6 Audit Trace: Fold 6 | Ctx: 20% | Gen: 100%

#	age	anaemia	creatinine_phosphoki	diabetes	ejection_fractio	high_blood_pressu	platelets	serum_creatinine	serum_sodium	sex
1	73.0	0.0	582.0	0.0	20.0	0.0	263358.03	1.83	134.0	1.0
2	80.0	0.0	582.0	1.0	35.0	0.0	350000.0	2.1	134.0	1.0
3	53.0	1.0	707.0	0.0	38.0	0.0	330000.0	1.4	137.0	1.0
4	45.0	0.0	7702.0	1.0	25.0	1.0	390000.0	1.0	139.0	1.0
5	60.0	1.0	257.0	1.0	30.0	0.0	150000.0	1.0	137.0	1.0
6	50.0	1.0	54.0	0.0	40.0	0.0	279000.0	0.8	141.0	1.0
7	81.0	0.0	4540.0	0.0	35.0	0.0	231000.0	1.18	137.0	1.0
8	70.0	0.0	2695.0	1.0	40.0	0.0	241000.0	1.0	137.0	1.0
9	70.0	0.0	835.0	0.0	35.0	1.0	305000.0	0.8	133.0	0.0
10	80.0	0.0	776.0	1.0	38.0	1.0	192000.0	1.3	135.0	0.0

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B.7 Audit Trace: Fold 7 | Ctx: 20% | Gen: 100%

#	age	anaemia	creatinine_phosphoki	diabetes	ejection_fractio	high_blood_pressu	platelets	serum_creatinine	serum_sodium	sex
1	80.0	0.0	898.0	0.0	25.0	0.0	149000.0	1.1	144.0	1.0
2	51.0	0.0	582.0	1.0	40.0	0.0	221000.0	0.9	134.0	0.0
3	60.0	1.0	582.0	0.0	30.0	1.0	127000.0	0.9	145.0	0.0
4	94.0	0.0	582.0	1.0	38.0	1.0	263358.03	1.83	134.0	1.0
5	70.0	1.0	75.0	0.0	35.0	0.0	223000.0	2.7	138.0	1.0
6	50.0	1.0	121.0	1.0	40.0	0.0	260000.0	0.7	130.0	1.0
7	75.0	0.0	582.0	1.0	30.0	1.0	263358.03	1.83	134.0	0.0
8	53.0	0.0	207.0	1.0	40.0	0.0	223000.0	1.2	130.0	0.0
9	60.0	1.0	2281.0	1.0	40.0	0.0	283000.0	1.0	141.0	0.0
10	59.0	1.0	280.0	1.0	25.0	1.0	302000.0	1.0	141.0	0.0

B.8 Audit Trace: Fold 8 | Ctx: 20% | Gen: 100%

#	age	anaemia	creatinine_phosphoki	diabetes	ejection_fractio	high_blood_pressu	platelets	serum_creatinine	serum_sodium	sex
1	53.0	0.0	63.0	1.0	60.0	0.0	368000.0	0.8	135.0	1.0
2	65.0	1.0	135.0	0.0	35.0	1.0	290000.0	0.8	134.0	1.0
3	50.0	1.0	249.0	1.0	35.0	1.0	319000.0	1.0	128.0	0.0
4	80.0	1.0	123.0	0.0	35.0	1.0	388000.0	9.4	133.0	1.0
5	60.0	0.0	59.0	0.0	25.0	1.0	212000.0	3.5	136.0	1.0
6	85.0	0.0	129.0	0.0	60.0	0.0	306000.0	1.2	132.0	1.0
7	65.0	0.0	224.0	1.0	50.0	0.0	149000.0	1.3	137.0	1.0
8	80.0	0.0	898.0	0.0	25.0	0.0	149000.0	1.1	144.0	1.0
9	66.0	1.0	72.0	0.0	40.0	1.0	242000.0	1.2	134.0	1.0
10	75.0	1.0	582.0	0.0	30.0	0.0	225000.0	1.83	134.0	1.0

B.9 Audit Trace: Fold 9 | Ctx: 20% | Gen: 100%

#	age	anaemia	creatinine_phosphoki	diabetes	ejection_fractio	high_blood_pressu	platelets	serum_creatinine	serum_sodium	sex
1	63.0	1.0	514.0	1.0	25.0	1.0	254000.0	1.3	134.0	1.0
2	65.0	1.0	305.0	0.0	25.0	0.0	298000.0	1.1	141.0	1.0
3	55.0	0.0	1199.0	0.0	20.0	0.0	263358.03	1.83	134.0	1.0
4	50.0	0.0	582.0	0.0	62.0	1.0	147000.0	0.8	140.0	1.0
5	73.0	1.0	231.0	1.0	30.0	0.0	160000.0	1.18	142.0	1.0
6	65.0	0.0	157.0	0.0	65.0	0.0	263358.03	1.5	138.0	0.0
7	55.0	0.0	572.0	1.0	35.0	0.0	231000.0	0.8	143.0	0.0
8	60.0	0.0	68.0	0.0	20.0	0.0	119000.0	2.9	127.0	1.0
9	70.0	0.0	97.0	0.0	60.0	1.0	220000.0	0.9	138.0	1.0
10	53.0	1.0	1808.0	0.0	60.0	1.0	249000.0	0.7	138.0	1.0

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B.10 Audit Trace: Fold 10 | Ctx: 20% | Gen: 100%

#	age	anaemia	creatinine_phosphoki	diabetes	ejection_fractio	high_blood_pressu	platelets	serum_creatinine	serum_sodium	sex
1	52.0	1.0	58.0	0.0	35.0	0.0	277000.0	1.4	136.0	0.0
2	82.0	1.0	379.0	0.0	50.0	0.0	47000.0	1.3	136.0	1.0
3	65.0	1.0	258.0	1.0	25.0	0.0	198000.0	1.4	129.0	1.0
4	65.0	0.0	395.0	1.0	25.0	0.0	265000.0	1.2	136.0	1.0
5	54.0	1.0	427.0	0.0	70.0	1.0	151000.0	9.0	137.0	0.0
6	58.0	1.0	400.0	0.0	40.0	0.0	164000.0	1.0	139.0	0.0
7	60.0	0.0	897.0	1.0	45.0	0.0	297000.0	1.0	133.0	1.0
8	75.0	0.0	582.0	0.0	20.0	1.0	265000.0	1.9	130.0	1.0
9	45.0	1.0	1876.0	1.0	35.0	0.0	226000.0	0.9	138.0	1.0
10	75.0	1.0	246.0	0.0	15.0	0.0	127000.0	1.2	137.0	1.0